SCIENCE & TECHNOLOGY PART 1 | CLASS 10 | MAHARASHTRA SSC BOARD

Chapter 8: Metallurgy

MODEL ANSWER SHEET / ANSWER KEY

165	24	23
QUESTIONS	IN TEXTBOOK	DIAGRAMS

HOW TO USE
Cover the answer box ✓ and answer first. A **TEXTBOOK** *strip means the same question is in your Balbharati Science-1 textbook, at the set and page shown. Tick ☐ each section once revised.*

01	02	03	04	05	06	07	08
MCQs 23 Q	Correlation 12 Q	Odd One Out 7 Q	Match Columns 6 Q	Write the Name 10 Q	Right or Wrong 12 Q	Identify who I am! 4 Q	Scientific Reasons 11 Q
09	**10**	**11**	**12**	**13**	**14**	**15**	**16**
Solve 20 Q	Distinguish 3 Q	Explain 13 Q	Explain 7 Q	Diagrams 6 Q	Complete flow chart gi 3 Q	Answer the Following 16 Q	Answer the Questions 12 Q

01 Choose the correct option.

23 QUESTIONS ☐ Revised

Q1 Bronze is an alloy of _____.

(a) **copper and tin** (b) copper and zinc
(c) tin and zinc (d) copper, zinc and tin

✓ **ANSWER / SOLUTION**
Bronze is an alloy of **copper and tin**.

Explanation:
Bronze is a solid alloy composed primarily of 88% copper and 12% tin. It is used in the production of hardware, utensils, musical instruments, medals, and other items. Due to its resistance to corrosion caused by seawater, it is used in submerged bearings and ship propellants.

Q2 _____ is an alloy made from iron, carbon and chromium.

(a) brass (b) bronze
(c) **stainless steel** (d) amalgam

✓ **ANSWER / SOLUTION**
Stainless steel is an alloy made from iron, carbon and chromium.

Q3 _____ is basic oxide.

(a) CO_2 (b) K_2O
(c) SO_2 (d) Al_2O_3

✓ **ANSWER / SOLUTION**
K_2O is basic oxide.

Q4 In electrolytic reduction of alumina _____ is used as a cathode.

(a) sulphur (b) **graphite**
(c) platinum (d) aluminium

✓ **ANSWER / SOLUTION**
In electrolytic reduction of alumina **graphite** is used as a cathode.

Q5 Iron is _____.

(a) more reactive than zinc (b) more reactive than aluminium
(c) less reactive than copper (d) **less reactive than aluminium**

✓ **ANSWER / SOLUTION**
Iron is **less reactive than aluminium**.

Q6 If Cu, Fe, Zn, Al elements are arranged in increasing order of their reactivity then the correct order would be which of the following?

(a) Cu, Fe, Zn, Al (b) Al, Cu, Fe, Zn
(c) Zn, Al, Cu, Fe (d) Fe, Zn, Al, Cu

ANSWER / SOLUTION
Cu, Fe, Zn, Al

Q7 Which of the following method is used to prevent the accumulation of greenish layer on brass due to corrosion?

(a) electroplating (b) anodization
(c) **tinning** (d) alloying

ANSWER / SOLUTION
tinning

Q8 In Wilfley table method to separate particles of gangue _____ method is used.

(a) Magnetic (b) Froth floatation
(c) Leaching (d) **gravitation**

ANSWER / SOLUTION
In Wilfley table method to separate particles of gangue **gravitation** method is used.

Q9 Aluminium oxide is _____ oxide.

(a) acidic (b) basic (c) neutral
(d) **amphoteric**

ANSWER / SOLUTION
Aluminium oxide is **amphoteric** oxide.

Q10 Atomic number of aluminium is _____ and its electronic configuration is _____.

(a) $13, (2, 8, 3)$ (b) $12, (2, 8, 2)$
(c) $13, (3, 10)$ (d) $12, (2, 10)$

ANSWER / SOLUTION
Atomic number of aluminium is 13 and its electronic configuration is $(2, 8, 3)$.

Q11 The chemical formula of zinc blend is _____.

(a) $ZnSO_4$ (b) ZnS
(c) $ZnCO_3$ (d) ZnO

ANSWER / SOLUTION
The chemical formula of zinc blend is ZnS.

Q12 Extraction of moderately reactive elements is done by _____ and _____ method.

(a) roasting and calcination (b) roasting and reduction
(c) separation and calcination (d) none of these

ANSWER / SOLUTION
Extraction of moderately reactive elements is done by **roasting** and **calcination** method.

Q13 Corrosion of silver causes a black layer of _____.

(a) Silver nitrate (b) silver oxide
(c) **silver sulphide** (d) silver carbonate

ANSWER / SOLUTION
Corrosion of silver causes a black layer of **silver sulphide**.

Q14 To prevent corrosion of iron and steel _____ method is used.

(a) electroplating (b) anodization
(c) tinning (d) **galvanizing**

ANSWER / SOLUTION
To prevent corrosion of iron and steel **galvanizing** method is used.

Q15 In preparation of Aqua regia hydrochloric acid and _____ acid are mixed.

(a) sulphuric acid (b) **nitric acid**
(c) carbonic acid (d) phosphoric acid

ANSWER / SOLUTION
In preparation of Aqua regia hydrochloric acid and **nitric acid** acid are mixed.

Q16 The sound of one metal colliding with another makes a noise, this property is called as _____.

(a) good conductors (b) ductility
(c) **sonority** (d) malleability

ANSWER / SOLUTION
The sound of one metal colliding with another makes a noise, this property is called as **sonority**.

Q17 Ionic compounds are electrically _____.

(a) positively charged (b) negatively charged
(c) **neutral** (d) conductor

ANSWER / SOLUTION
Ionic compounds are electrically **neutral**.

Q18 _____ is good conductor of heat but bad conductor of electricity.

(a) graphite (b) **diamond** (c) coal (d) iodine

ANSWER / SOLUTION
Diamond is good conductor of heat but bad conductor of electricity.

Q19 _____ is the least reactive metal.

(a) silver (b) sodium (c) zinc (d) **gold**

ANSWER / SOLUTION
Gold is the least reactive metal.

Q20 _____ forms a green colour in the water.

(a) $CuSO_4$ (b) $FeSO_4$
(c) $NaCl$ (d) all the above

ANSWER / SOLUTION
$FeSO_4$ forms a green colour in the water.

Q21 Stainless steel is an alloy of _____.

(a) copper (b) tin (c) zinc (d) **iron**

ANSWER / SOLUTION
Stainless steel is an alloy of **iron**.

Q22 When one of the metals in an alloy is mercury the alloy is called _____.

(a) **amalgam** (b) sodium amalgam
(c) zinc amalgam (d) all the above

ANSWER / SOLUTION
When one of the metals in an alloy is mercury the alloy is called **amalgam**.

Q23 The minerals from which the metal can be separated economically are called _____.

(a) minerals (b) **ores** (c) gangue (d) alloy

ANSWER / SOLUTION
The minerals from which the metal can be separated economically are called **ores**.

02 Find the correlation.

12 QUESTIONS ☐ Revised

Q1 Brass : Aluminium and zinc : : Bronze : _____

ANSWER / SOLUTION
Brass : Aluminium and zinc : : Bronze : **Copper and tin**

Q2 Pressure cooker : Anodizing : : Silver plated spoons : _____

ANSWER / SOLUTION
Pressure cooker : Anodizing : : Silver plated spoons : **Electroplating**

Q3 In Electrolytic reduction of alumina, Anode : _____ : : Cathode : Graphite lining

ANSWER / SOLUTION
In Electrolytic reduction of alumina, Anode : **Set of carbon (graphite) rods** : : Cathode : Graphite lining

Q4 Sulphide ores : Roasting : : Carbonate ores : _____

ANSWER / SOLUTION
Sulphide ores : Roasting : : Carbonate ores : **Calcination**

Q5 Bauxite : Aluminium ore : : Cassiterite : _____

ANSWER / SOLUTION
Bauxite : Aluminium ore : : Cassiterite : **Tin**

Q6 Metal sheets : Malleable : : Electric wires : _____

ANSWER / SOLUTION
Metal sheets : Malleable : : Electric wires : **Ductile**

Q7 Zinc sulphide : Roasting : : zinc carbonate : _____

ANSWER / SOLUTION

Zinc sulphide : Roasting : : zinc carbonate : **calcination**

Q8 Rusting of iron : Fe_2O_3 : : corrosion of copper : _____

ANSWER / SOLUTION

Rusting of iron : Fe_2O_3 : : corrosion of copper : $CuCO_3$

Q9 Diamond : electric insulator : : _____ : electric conductor

ANSWER / SOLUTION

Diamond : electric insulator : : **graphite** : electric conductor

Q10 Soft metal : Na : : hard metal : _____

ANSWER / SOLUTION

Soft metal : Na : : hard metal : **Tungsten**

Q11 Aluminium: _____: gold : : electric insulator

ANSWER / SOLUTION

Aluminium: **electric conductor** : gold : : electric insulator

Q12 Bronze : _____ : : Stainless steel : $Fe + Cr + C$

ANSWER / SOLUTION

Bronze : $Cu + Sn$: : Stainless steel : $Fe + Cr + C$

03 Find odd one out.

7 QUESTIONS · Revised

Q1 Find the odd one out and give its explanation.

(a) Sodium (b) Potassium (c) Silver (d) **Sulphur**

ANSWER / SOLUTION

Sulphur

Explanation-

Sulphur is a nonmetal while others are metals.

Q2 Find the odd one out and give its explanation.

(a) **Boron** (b) Chlorine (c) Bromine (d) Fluorine

ANSWER / SOLUTION

Boron

Explanation-

Boron is placed in group 13 while others are placed in group 17 of the modern periodic table.

Q3 Find the odd one out and give its explanation.

(a) Copper (b) Iron (c) Mercury (d) **Brass**

ANSWER / SOLUTION

Brass

Explanation-

Brass is an alloy while others are pure metals.

Q4 Find the odd one out and give its explanation.

(a) Brass (b) Bronze (c) **Phosphorous** (d) Steel

ANSWER / SOLUTION

Phosphorous

Explanation-

Phosphorus is a nonmetal while others are alloys.

Q5 Find the odd one out and give its explanation.

(a) Tinning (b) Alloying (c) Anodization (d) **Froth floatation**

ANSWER / SOLUTION

Froth floatation

Explanation-

Froth floatation is a method used for the concentration of ores while others are methods used for the prevention of corrosion.

Q6 **Find the odd one out and give its explanation.**

(a) Zinc coating (b) Tinning
(c) Electroplating (d) **Calcination**

ANSWER / SOLUTION

Calcination

Explanation-

Calcination is a process involved in extraction of carbonate ores while others are methods used for prevention of corrosion.

Q7 **Find the odd one out and give its explanation.**

(a) Na (b) K
(c) Cu (d) Li

ANSWER / SOLUTION

Cu

Explanation-

Copper (Cu) is a less reactive metal while others are highly reactive metals.

04 Match the columns

6 QUESTIONS · 2 · Revised

Q1 **Make pairs of substances and their properties**

Substance	Property
a. Potassium bromide (KBr)	1. Combustible
b. Gold	2. Soluble in water
c. Sulphur	3. No chemical reaction
d. Neon	4. High ductility.

TEXTBOOK Exercise · Q.2 · Page 108

ANSWER / SOLUTION

Substance	Property
a. Potassium bromide (KBr)	2. Soluble in water
b. Gold	4. High ductility
c. Sulphur	1. Combustible
d. Neon	3. No chemical reaction

Q2 **Identify the pairs of metals and their ores from the following.**

Group A	Group B
a. Bauxite	i. Mercury
b. Cassiterite	ii. Aluminium
c. Cinnabar	iii. Tin

TEXTBOOK Exercise · Q.3 · Page 109

ANSWER / SOLUTION

Group A	Group B
a. Bauxite	ii. Aluminium
b. Cassiterite	iii. Tin
c. Cinnabar	i. Mercury

Q3 **Match the columns.**

Group A	Group B
1. ZnS	a) Copper Sulphide
2. HgS	b) Bauxite
	c) Cinnabar
	d) Zinc blend

ANSWER / SOLUTION

Group A	Group B
1. ZnS	d) Zinc blend
2. HgS	c) Cinnabar

Q4 **Match the columns.**

Group A	Group B
1) Copper and Zinc	a) Brass
2) Copper and Tin	b) Bronze
	c) Stainless steel

ANSWER / SOLUTION

Group A	Group B
1) Copper and Zinc	a) Brass
2) Copper and Tin	b) Bronze

Q5 **Match the columns.**

Group A	Group B
1) Electroplating	a) Pressure cooker
2) Anodising	b) Silver plated spoons
	c) Coating of tin on copper
	d) Coating of Zinc on iron

ANSWER / SOLUTION

Group A	Group B
1) Electroplating	b) Silver plated spoons
2) Anodising	a) Pressure cooker

Q6 **Match the columns.**

Group A	Group B
1. Making sheets of metals	a) Sonority
2. Making metal utensils	b) Malleability
3. Making Copper wires	c) Good conductor of heat
4. Making bells from metal	d) Ductility

ANSWER / SOLUTION

Group A	Group B
1. Making sheets of metals	b) Malleability
2. Making metal utensils	c) Good conductor of heat
3. Making Copper wires	d) Ductility
4. Making bells from metal	a) Sonority

05 Write the name

10 QUESTIONS · 4 · Revised

Q1 **Write the name.**
The molecular formula of main ore of aluminium –

TEXTBOOK Exercise · Q.1 (b) · Page 108

ANSWER / SOLUTION
The molecular formula of main ore of aluminium – $Al_2O_3 \cdot nH_2O$

Q2 **Write name.**
The device used for grinding an ore.

TEXTBOOK Exercise · Q.1 (d) · Page 108

ANSWER / SOLUTION
The device used for grinding an ore- **Ball mill**

Q3 **Write the name.**
Nonmetal which is a good conductors of electricity.

TEXTBOOK Exercise · Q.1 (e) · Page 108

ANSWER / SOLUTION
Nonmetal which is a good conductors of electricity- **Graphite**

Q4 **Write the name.**
The reagent which dissolves noble metals.

TEXTBOOK Exercise · Q.1 (f) · Page 108

ANSWER / SOLUTION
The reagent which dissolves noble metals- **Aqua regia**

Q5 **Write the name.**
Metals which are amphoteric in nature.

ANSWER / SOLUTION
Metals which are amphoteric in nature- **Aluminium** (Al) **, zinc** (Zn)

Q6 **Write the name.**
An alloy of copper and tin-

ANSWER / SOLUTION
An alloy of copper and tin- **Bronze**

Q7 **Write the name.**
Two highly reactive metals-

ANSWER / SOLUTION
Two highly reactive metals- **Lithium** (Li)**, sodium** (Na)

Q8 **Write the name.**
The process of strong heating of carbonate ores in insufficient air–

✓ **ANSWER / SOLUTION**
The process of strong heating of carbonate ores in insufficient air– **Calcination**

Q9 **Write the name.**
The process of extraction of aluminium from alumina-

✓ **ANSWER / SOLUTION**
The process of extraction of aluminium from alumina- **Electrolytic reduction**

Q10 **Write the name.**
Method used to prevent corrosion of copper.

✓ **ANSWER / SOLUTION**
Method used to prevent corrosion of copper- **Anodization and/or tinning**

06 Right or Wrong sentence

12 QUESTIONS ☐ Revised

Q1 **Electrolysis is used to obtain pure metal from impure metal.**
(a) **Right** (b) Wrong

✓ **ANSWER / SOLUTION**
Electrolysis is used to obtain pure metal from impure metal- **Right**

Q2 **Ionic compounds are soluble in kerosene.**
(a) Right (b) **Wrong**

✓ **ANSWER / SOLUTION**
Ionic compounds are soluble in kerosene- **Wrong**

Q3 **In solid state, ionic compounds conduct electricity.**
(a) Right (b) **Wrong**

✓ **ANSWER / SOLUTION**
In solid state, ionic compounds conduct electricity- **Wrong**

Q4 **Mercury, silver, gold are highly reactive metals.**
(a) Right (b) **Wrong**

✓ **ANSWER / SOLUTION**
Mercury, silver, gold are highly reactive metals- **Wrong**

Q5 **In the electrolytic method, a layer of highly active metal is applied to a less active metal.**
(a) Right (b) **Wrong**

✓ **ANSWER / SOLUTION**
In the electrolytic method, a layer of highly active metal is applied to a less active metal- **Wrong**

Q6 **In the electrolytic reduction of alumina, the lining of graphite acts as an anode.**
(a) Right (b) **Wrong**

✓ **ANSWER / SOLUTION**
In the electrolytic reduction of alumina, the lining of graphite acts as an anode- **Wrong**

Q7 **The electrolysis of alumina involves the use of fluorspar and cryolite to increase the melting point.**
(a) Right (b) **Wrong**

✓ **ANSWER / SOLUTION**
The electrolysis of alumina involves the use of fluorspar and cryolite to increase the melting point- **Wrong**

Q8 **Cassiterite is a copper ore.**
(a) Right (b) **Wrong**

✓ **ANSWER / SOLUTION**
Cassiterite is a copper ore- **Wrong**

Q9 Diamond is a hard substance.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Diamond is a hard substance- **Right**

Q10 Gold and silver are active metals.

(a) Right (b) **Wrong**

ANSWER / SOLUTION

Gold and silver are active metals- **Wrong**

Q11 Halogen reacts with acid.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Halogen reacts with acid- **Right**

Q12 Bauxite reacts with sodium hydroxide in the Bayer's process.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Bauxite reacts with sodium hydroxide in the Bayer's process- **Right**

07 Identify who I am!

4 QUESTIONS ☐ Revised

Q1 Identify who I am!
Carbon allotropes- _____

ANSWER / SOLUTION

Carbon allotropes- **Diamond and graphite**

Q2 Identify who I am!
Amphoteric oxide forming metal- _____

ANSWER / SOLUTION

Amphoteric oxide forming metal- **Aluminium**

Q3 Identify who I am!
Ore of Aluminum- _____

ANSWER / SOLUTION

Ore of Aluminum- **Bauxite**

Q4 Identify who I am!
Metal in Liquid state- _____

ANSWER / SOLUTION

Metal in Liquid state- **Mercury**

08 Write scientific reasons

11 QUESTIONS · 6 ☐ Revised

Q1 **Write scientific reason.**
Sodium is always kept in kerosene.

TEXTBOOK Exercise · Q.5 (c) · Page 109

ANSWER / SOLUTION

Kerosene contains sodium as:

1. As a highly reactive metal, sodium interacts with water in a highly exothermic manner.
2. It has a strong reaction with water and oxygen.
3. It can generate much heat when it reacts with oxygen and water vapor in the atmosphere.
4. To prevent this strong reaction, it is thus stored or maintained in kerosene.

Solution 2 (Alternative):

1. Sodium reacts with moisture present in air, even at room temperature, to form sodium hydroxide and hydrogen gas.

$$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)} \uparrow$$

2. The reaction is exothermic and therefore, the released hydrogen gas readily catches fire on exposure to air. Hence, to prevent accidents, sodium is stored in kerosene.

Q2 **Write scientific reason.**
Pine oil is used in froth flotation.

TEXTBOOK Exercise · Q.5 (d) · Page 109

ANSWER / SOLUTION

1. Pine oil is hydrophobic.
2. The particles of the metal sulphide ores which are hydrophobic get wetted by pine oil.
3. They float with the foam on the surface of water.
4. Thus, by using this property some ores which contain hydrophobic particles in them can be concentrated by froth floatation process.
   Hence, pine oil is used in froth floatation.

Q3 **Write scientific reasons.**
Lemon or tamarind is used for cleaning copper vessels turned greenish.

TEXTBOOK Exercise · Q.5 (a) · Page 109

ANSWER / SOLUTION

1. Copper vessels turned greenish due to the formation of copper carbonate layer.
2. The citric acid present in the lemon or tamarind neutralizes the basic copper carbonate and dissolves the layer.
3. That is why tarnished copper vessels are cleaned with lemon or tamarind juice to give the surface of the copper vessel its characteristic lustre.

Q4 **Write scientific reason.**
Anodes need to be replaced from time to time during the electrolysis of alumina.

TEXTBOOK Exercise · Q.5 (e) · Page 109

ANSWER / SOLUTION

In the electrolysis of alumina, graphite rod is used as anode.

- During the electrolytic reduction of alumina, aluminium is produced at the cathode and oxygen gas is evolved at the anode.
- This gas reacts with the graphite rods (anode) and forms carbon dioxide.

Thus, the graphite rods are continuously eroded. Hence, anodes are replaced from time to time during the electrolysis of alumina.

Q5 **Write scientific reason.**
Generally the ionic compounds have high melting points.

TEXTBOOK Exercise · Q.5 (b) · Page 109

ANSWER / SOLUTION

1. Ionic compounds are composed of cations and anions.
2. These ions are held together by strong electrostatic forces of attraction.
3. Considerable amount of energy is required to break these strong forces of attraction.
   Hence, the melting points of ionic compounds are high.

Q6 **Write scientific reason.**
Anodization method is useful for prevention of the corrosion of the aluminium.

ANSWER / SOLUTION

1. The process of coating metals such as aluminium and copper with a thin and strong layer of their oxides by electrolysis is called anodizing process.
2. When aluminium is anodized, a thin layer of aluminium oxide is formed all over the surface. This restricts the contact of aluminium metal with oxygen and water and prevents corrosion of aluminium. Hence, anodization method is useful for prevention of the corrosion of the aluminium.

Q7 **Write scientific reason.**
On exposure to air, silver articles turn blackish after some time.

ANSWER / SOLUTION

1. Silver articles turn blackish after some time of exposure to air. This is because silver undergoes corrosion when exposed to air containing hydrogen sulphide.
2. Silver reacts with hydrogen sulphide in air to form a layer of silver sulphide (Ag_2S), which gives silver articles a blackish appearance.

Q8 **Write scientific reason.**
Magnetic separation method is used to separate the magnetic ingredients in the ores.

ANSWER / SOLUTION

1. The magnetic separation process is based on the differences in magnetic properties of the ore components.
2. When the powdered ore is dropped over the moving belt near nonmagnetic roller, the nonmagnetic particles are carried further along with belt and fall in the collector vessel placed away from magnetic roller. At the same time, the magnetic portion of the ore is attracted by the magnetic roller and falls in the collector vessel near the magnetic roller.
   Thus, magnetic separation method is used to separate the magnetic ingredients in the ores.

Q9 **Write scientific reason.**
Coins are made from metals and alloys.

ANSWER / SOLUTION

1. Coins are made from metals and alloys because metals and their alloys possess the property of malleability. So, they can be hammered into flat shape without breaking.
2. Metals and alloys used for making coins should be corrosion resistance and bright in colour. They should also be hard and wear resistance.

Q10 **Write scientific reason.**
Meena's mother uses lemon or tamarind for cleaning copper vessels turned greenish.

ANSWER / SOLUTION

1. The greenish colour on copper vessels is due to the formation of copper carbonate on it.
2. Copper carbonate reacts with acids present in lemon or tamarind and as a result, this copper carbonate layer gets removed.
   Hence, Meena's mother uses lemon or tamarind for cleaning copper vessels turned greenish.

Q11 **Write scientific reason.**
Sodium is always kept in kerosene.

TEXTBOOK **Exercise · Q.5 (c) · Page 109**

ANSWER / SOLUTION

1. Sodium reacts with moisture present in air, even at room temperature, to form sodium hydroxide and hydrogen gas.
   $$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)} \uparrow$$
2. The reaction is exothermic and therefore, the released hydrogen gas readily catches fire on exposure to air. Hence, to prevent accidents, sodium is stored in kerosene.

09 Solve the following Questions

20 QUESTIONS · 2 | Revised

Q1 When a copper coin is dipped in the solution of silver nitrate, it shows a lustre on its surface. Explain this process with the help of a chemical equation.

TEXTBOOK Exercise · Q.6 · Page 109

ANSWER / SOLUTION

1. Copper is more reactive than silver. Therefore, copper displaces silver from its salt solution, silver nitrate. The glittering white deposit of silver metal is formed on the copper coin. Hence, when a copper coin is dipped in the solution of silver nitrate, it shows a lustre on its surface after some time.
2. The reaction is as follows:

$$\mathrm{Cu}_{(s)} + 2\mathrm{AgNO}_{3(aq)} \rightarrow \mathrm{Cu(NO_3)}_{2(aq)} + 2\mathrm{Ag}_{(s)}$$

Q2 Electronic configuration of metal A is $2, 8, 1$. Electronic configuration of metal B is $2, 8, 8, 2$. Which of the above metals is more reactive? Explain with reason.

ANSWER / SOLUTION

1. Metal A with electronic configuration of $(2, 8, 1)$ is sodium (Na).
2. Metal B with electronic configuration of $(2, 8, 8, 2)$ is calcium (Ca).
3. Metal A is more reactive than B as it has to lose only one electron from the outermost shell while metal B has to lose two electrons, to achieve a stable electronic configuration.

Q3 Classify the following metals based on their reactivity. Cu, Zn, Ca, Mg, Fe, Na, Li, Hg

ANSWER / SOLUTION

More reactive	Moderately reactive	Less reactive
Ca	Zn	Cu
Na	Mg	Hg
Li	Fe	

Q4 Write the molecular formulae of the following compound.
Cryolite

ANSWER / SOLUTION

Cryolite- $\mathrm{Na_3AlF_6}$

Q5 Write the molecular formulae of the following compound.
Fluorspar

ANSWER / SOLUTION

Fluorspar- $\mathrm{CaF_2}$

Q6 Write the molecular formulae of the following compound.
Sodium aluminate

ANSWER / SOLUTION

Sodium aluminate- $\mathrm{NaAlO_2}$

Q7 Write the molecular formulae of the following compound.
Copper pyrite

ANSWER / SOLUTION

Copper pyrite- $\mathrm{CuFeS_2}$

Q8 Write the molecular formulae of the following compound.
Stannic oxide

ANSWER / SOLUTION

Stannic oxide- $\mathrm{SnO_2}$

Q9 Write the molecular formulae of the following compound.
Ferrous tungstate

ANSWER / SOLUTION

Ferrous tungstate- $\mathrm{FeWO_4}$

Q10 Explain the concept of Roasting.

ANSWER / SOLUTION

In roasting process, the sulphide ores are strongly heated in air to transform them into oxides. In this process, sulphide ores are oxidized to form metal oxides and sulphur dioxide gas.

Q11 Explain the concept of Calcination.

ANSWER / SOLUTION

In the calcination process, carbonate ores are strongly heated in a limited supply of air to transform them into oxides. In this process, carbonate ores decompose to form a metal oxide and carbon dioxide gas.

Q12 What is an alloy?

ANSWER / SOLUTION

The homogeneous mixture formed by mixing a metal with other metals or nonmetals in certain proportion is called an alloy.

Q13 Give two examples of alloys.

ANSWER / SOLUTION

1. Bronze
2. Stainless steel

Q14 Explain Bayer's process.

ANSWER / SOLUTION

1. In Bayer's process, the bauxite ore is first crushed and then it is leached by heating with hot concentrated caustic soda ($NaOH$) solution under high pressure for 2 to 8 hrs at 140 to $150\,^{\circ}C$ in a tank called a digester.
2. Aluminium oxide, being amphoteric in nature, dissolves in an aqueous $NaOH$ solution, forming water-soluble sodium aluminate.
$$Al_2O_3 \cdot 2H_2O_{(s)} + 2NaOH_{(aq)} \rightarrow 2NaAlO_{2(aq)} + 3H_2O_{(l)}$$
3. The iron oxide present in the gangue does not dissolve in aqueous $NaOH$ solution and is separated by filtration.
4. However, silica from the gangue dissolves in aqueous $NaOH$ solution forming water-soluble sodium silicate.
5. The solution of sodium aluminate is diluted with water and then cooled to $50\,^{\circ}C$. This gives aluminium hydroxide as a precipitate.
$$NaAlO_{2(aq)} + 2H_2O_{(l)} \rightarrow NaOH_{(aq)} + Al(OH)_3 \downarrow$$
6. The aluminium hydroxide is then filtered, washed, dried and calcined by heating at $1000\,^{\circ}C$ to get pure aluminium oxide, called alumina.
$$2Al(OH)_{3(s)} \xrightarrow[1000\,^{\circ}C]{\text{heat}} Al_2O_{3(s)} + 3H_2O_{(g)}$$

Q15 What is the difference between calcination and roasting?

ANSWER / SOLUTION

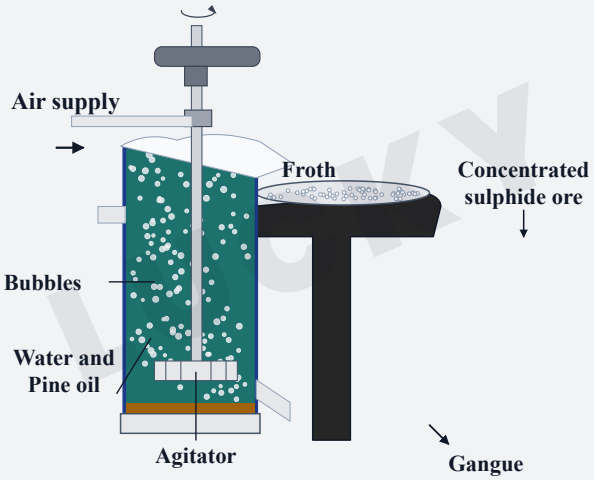
#	Roasting	Calcination
(1)	The ore is heated in the excess of air.	The ore is heated in the absence of air.
(2)	Volatile impurities are removed as oxides (SO_2, P_2O_5, As_2O_3) and the ore becomes porous and more reactive.	Moisture and organic impurities are removed and the ore becomes porous and more reactive.
(3)	Sulphide ores are roasted, so SO_2 is given off. $2ZnS + 3O_2 \xrightarrow{800^{\circ}-900^{\circ}C} 2ZnO + 2SO_2$	Carbonate and hydrated ores are calcined and CO_2 or water vapour are given off. $ZnCO_3 \rightarrow ZnO + CO_2$

Q16 Draw a neat and labelled diagram for Froth Floatation Process.

TEXTBOOK Exercise · Q.8 (b) · Page 109

ANSWER / SOLUTION

Froth floatation method-



Q17 Explain the reactions of nonmetals with water with the help of examples.

ANSWER / SOLUTION

Generally, nonmetals do not react with water, except halogens.

E.g. Chlorine on dissolving in water gives the following reaction:
$$Cl_{2(g)} + H_2O_{(l)} \rightarrow HOCl_{(aq)} + HCl_{(aq)}$$

Q18 Explain the characteristics of ionic compounds.

ANSWER / SOLUTION

1. **Physical nature:** Ionic compounds are solids and are hard. They are generally brittle and break into pieces when pressure is applied.
2. **Melting and boiling points:** Ionic compounds have high melting and boiling points.
3. **Solubility:** Ionic compounds are generally soluble in water. However, they are insoluble in kerosene and petrol.
4. **Electrical conductivity:** Ionic compounds conduct electricity in a molten state or in its aqueous solution. Conduction of electricity in these states is due to the movement of charged particles (i.e., ions) towards the oppositely charged electrodes. However, in solid-state, ionic compounds do not conduct electricity as movement of the ions is not possible due to their rigid structure.

Q19 What are amphoteric oxides?

ANSWER / SOLUTION

Those metal oxides which show basic as well as acidic behaviour are known as amphoteric oxides.

Q20 Give two examples of amphoteric oxides.

ANSWER / SOLUTION

1. Aluminium oxide (Al_2O_3)
2. Zinc oxide (ZnO)

10 Distinguish between

3 QUESTIONS ☐ Revised

Q1 **Distinguish between:**
Metals - Nonmetals (physical characteristics)

ANSWER / SOLUTION

#	Metals	Nonmetals
1.	Metals are generally solids at room temperature. **Exception:** Mercury is liquid at room temperature.	Nonmetals are generally gases or solids at room temperature. **Exception:** Bromine is in liquid state at room temperature.
2.	Metals are hard. **Exceptions:** Sodium, potassium.	Nonmetals are generally soft. **Exception:** Diamond
3.	Metals have high melting and boiling points. **Exceptions:** Sodium, potassium, mercury, gallium.	Nonmetals have low melting and boiling points.
4.	Metals have lustre.	Nonmetals have no lustre. **Exception:** Iodine
5.	Metals are good conductors of heat and electricity.	Nonmetals are bad conductors of heat and electricity. **Exception:** Graphite is good conductor of electricity.
6.	Most of the metals are sonorous i.e., they produce a sound when stuck on hard surface.	Nonmetals are not sonorous.
7.	Metals are malleable and ductile.	Nonmetals are not malleable and ductile.

Q2 Compare roasting and calcination.

ANSWER / SOLUTION

#	Roasting	Calcination
1.	This method is generally used for sulphide ores	This method is generally used for carbonate ores.
2.	In this method, ores are heated in excess of air.	In this method, ores are heated in a limited supply of air.
3.	In roasting, generally sulphur dioxide gas is evolved.	In calcination, generally carbon dioxide gas is evolved.
4.	In this method, sulphide ores are oxidized to form metal oxides. **E.g.** $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2 \uparrow$	In this method, carbonate ores decompose to form metal oxide. **E.g.** $ZnCO_3 \rightarrow ZnO + CO_2 \uparrow$

Q3 **Distinguish between:**
Froth floatation - Leaching

ANSWER / SOLUTION

#	Froth floatation	Leaching
1.	The froth floatation method is based on the two opposite properties, hydrophilic and hydrophobic, of the gangue and ore particles respectively.	Leaching method is based on the dissolution of ore particles in a certain solution.
2.	There is no chemical reaction involved.	There is a specific chemical reaction between ore and the solution used for leaching.
3.	This method involves the formation of froth from oil, water, and air bubbles together, due to the agitation.	There is no formation of froth.
4.	This method is used for concentration of zinc blende and copper pyrite.	This method is used for the concentration of bauxite.

11 Explain the following reactions with the balanced equations

13 QUESTIONS · 3 ☐ Revised

Q1 **Explain the following reaction with the balanced equation.**
Sodium burns in air

ANSWER / SOLUTION

On burning, sodium metal combines with oxygen in the air to form sodium oxide.

$$4Na_{(s)} + O_{2(g)} \rightarrow 2Na_2O_{(s)}$$

Q2 **Explain the following reaction with the balanced equation.**
Reaction of aluminium with oxygen

ANSWER / SOLUTION

Aluminium combines with oxygen to form aluminium oxide.

$$4Al_{(s)} + 3O_{2(g)} \rightarrow 2Al_2O_{3(s)}$$

Q3 **Explain the following reaction with the balanced equation.**
Magnesium reacts with dil HCl

ANSWER / SOLUTION

When magnesium is treated with dilute hydrochloric acid, it forms magnesium chloride with release of hydrogen gas.

$$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)} \uparrow$$

Q4 **Explain the following reaction with the balanced equation.**
Sulphur burns in air

ANSWER / SOLUTION

Sulphur burns in air to form sulphur dioxide.

$$S_{(s)} + O_{2(g)} \rightarrow SO_{2(g)}$$

Q5 **Explain the following reaction with the balanced equation.**
Chlorine dissolved in water

ANSWER / SOLUTION

Chlorine dissolves in water to form hypochlorous acid and hydrochloric acid.

$$Cl_{2(g)} + H_2O_{(l)} \rightarrow HOCl_{(aq)} + HCl_{(aq)}$$

Q6 **Explain the following reaction with the balanced equation.**
Sodium aluminate reacts with water

ANSWER / SOLUTION

Sodium aluminate reacts with water to give aluminium hydroxide as a precipitate and sodium hydroxide.

$$NaAlO_{2(aq)} + 2H_2O_{(l)} \rightarrow NaOH_{(aq)} + Al(OH)_3 \downarrow$$

Q7 **Write a chemical equation for the event.**
Iron filings are dropped in aqueous solution of copper sulphate.

TEXTBOOK **Exercise · Q.9 (b) · Page 109**

ANSWER / SOLUTION

More reactive iron pushes copper out of the copper sulphate solution when iron filings are added. Reddish-brown copper metal coats the iron filings, and ferrous sulphate is created as the blue colour of copper sulphate gradually disappears.

$$\underset{\text{(Blue)}}{Fe_{(s)} + CuSO_{4(aq)}} \rightarrow \underset{\text{(Colourless)}}{FeSO_{4(aq)}} + Cu_{(s)}$$

Q8 **Explain the following reaction with the balanced equation.**
Ferric oxide is reacted with aluminium.

TEXTBOOK **Exercise · Q.9 (c) · Page 109**

ANSWER / SOLUTION

Ferric oxide reacts with aluminium to form iron and aluminium oxide. This reaction is known as the thermit reaction.

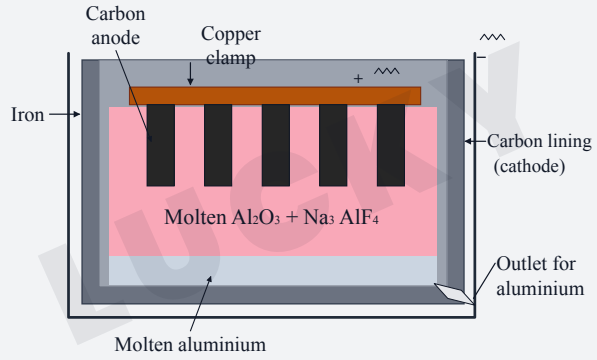
$$Fe_2O_{3(s)} + 2Al_{(s)} \xrightarrow{\text{Ignited}} 2Fe_{(l)} + Al_2O_{3(s)} + \text{Heat}$$

Q9 **Write chemical equation for the event.**
Electrolysis of alumina is done.

TEXTBOOK **Exercise · Q.9 (d) · Page 109**

ANSWER / SOLUTION

The electrolysis of alumina is carried out in a steel tank lined inside with graphite. The graphite lining serves as cathode. Anode is also made up of graphite rods hanging in the molten mass. The electrolyte consists of alumina dissolved in fused Cryolite (Na_3AlF_6) and Fluorspar (CaF_2). Cryolite lowers the melting point of alumina and fluorspar increases the fluidity of the mass so that the liberated aluminum metal may sink at the bottom of the cell. When electric current is passed through this mixture, the aluminum is collected at the cathode in molten state and sinks at the bottom.



Electrolytic cell for the extraction of aluminium

Ionization of Alumina: $2Al_2O_3 \rightarrow 6O^{2-} + 4Al^{3+}$
Reaction at Cathode: $4Al^{3+} + 12e^- \rightarrow 4Al$
Reaction at Anode: $6O^{2-} \rightarrow 3O_2 + 12e^-$, $C + O_2 \rightarrow CO_2$

Solution 2 (Alternative):
When electrolysis of alumina is done, aluminium is formed at the cathode and oxygen gas liberated at the anode.

Cathode: $Al^{3+} + 3e^- \rightarrow Al_{(l)}$ (Reduction)

Anode: $2O^{2-} \rightarrow O_{2(g)} + 4e^-$ (Oxidation)

Q10 **Explain the following reaction with the balanced equation.**
Dry aluminium hydroxide is ignited at $1000\,^\circ C$

ANSWER / SOLUTION

When dry aluminium hydroxide is calcined at $1000\,^\circ C$, pure aluminium oxide, called alumina is obtained.

$$2Al(OH)_{3(s)} \xrightarrow[1000\,^\circ C]{\text{heat}} Al_2O_{3(s)} + 3H_2O_{(g)}$$

Q11 **Explain the following reaction with the balanced equation.**
Zinc sulphide is heated strongly in excess of air

ANSWER / SOLUTION

Zinc sulphide is heated strongly in excess of air to convert it to zinc oxide. This process is called roasting.

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_2 \uparrow$$

Q12 **Explain the following reaction with the balanced equation.**
Zinc carbonate is heated strongly in limited supply of air

ANSWER / SOLUTION

Zinc carbonate is heated strongly in a limited supply of air to convert it to zinc oxide. This process is called calcination.

$$ZnCO_{3(s)} \xrightarrow{\text{Heat}} ZnO_{(s)} + CO_2 \uparrow$$

Q13 **Explain the following reaction with the balanced equation.**
Zinc oxide is treated with carbon

ANSWER / SOLUTION

Zinc oxide is reduced to zinc by heating it with carbon.

$$ZnO_{(s)} + C_{(s)} \xrightarrow{\text{Heat}} Zn_{(s)} + CO \uparrow$$

12 Explain concept with example/explain with the help of balanced equation

7 QUESTIONS · 3 · Revised

Q1 **Explain concept with example/explain with the help of a balanced equation.**
Ionic bond and ionic compounds

ANSWER / SOLUTION

1. The compounds formed from two units, namely cation and anion are called ionic compounds.
2. The cation and anion being oppositely charged, there is an electrostatic force of attraction between them. This force of attraction between cation and anion is called as the ionic bond.
3. The number of cations and anions in a compound and the magnitude of the electric charge on them is such that the positive and negative charges balance each other. As a result, an ionic compound is electrically neutral.
4. For example, ionic compound sodium chloride ($NaCl$) is formed as sodium metal gives away one electron while the nonmetal chlorine takes up one electron. Na^+ and Cl^- ions formed are held together by attractive force called ionic bond.

Q2 **Explain concept with example/explain with the help of a balanced equation.**
Gangue

TEXTBOOK Exercise · Q.4 (d) · Page 109

ANSWER / SOLUTION

1. The unwanted impurities of sand, soil, rocky substances, etc., present in the ore are called gangue.
2. Ores are taken out from the mines and the gangue is usually separated from the ore at the site itself by various methods.
3. The process of separating gangue from the ores is called the concentration of ores.
4. For example, bauxite ore contains impurities of silica (SiO_2), ferric oxide (Fe_2O_3) and titanium oxide (TiO_2). These impurities are called gangue.

Q3 **Explain concept with example/explain with the help of a balanced equation.**
Ores

TEXTBOOK Exercise · Q.4 (b) · Page 109

ANSWER / SOLUTION

1. There can be many minerals from which a metal can be extracted.
2. However, only those minerals from which a metal can be extracted profitably are called ores.
3. Metals can be extracted from their ores by means of various methods of separation.
4. For example, aluminium can be extracted profitably from its bauxite ore. Hence, bauxite is an ore of aluminium.

Q4 **Explain concept with example/explain with the help of a balanced equation.**
Roasting

ANSWER / SOLUTION

1. The process of heating of sulphide ores strongly in excess of air to convert it into respective oxide is called roasting.
2. For example, zinc sulphide is heated strongly in excess of air to convert it to zinc oxide.

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_2 \uparrow$$

Q5 **Explain concept with example/explain with the help of a balanced equation.**
Calcination

ANSWER / SOLUTION

1. The process of heating carbonate ores in limited air to convert it into respective oxide is called calcination.
2. For example, zinc carbonate is heated strongly in a limited supply of air to convert it to zinc oxide.

$$ZnCO_{3(s)} \xrightarrow{\text{Heat}} ZnO_{(s)} + CO_2 \uparrow$$

Q6 **Explain concept with example/explain with the help of a balanced equation.**
Corrosion

ANSWER / SOLUTION

1. The process of slow degradation of the metals due to the attack of atmospheric gases, moisture, acids, etc. on the surface of metals is called corrosion.
2. Iron reacts with moist air and a deposit of reddish substance ($Fe_2O_3 \cdot H_2O$) is formed on it. This substance is called rust and the corrosion of iron is known as rusting.
3. Carbon dioxide in moist air reacts with the surface of copper vessel. Copper loses its lustre due to formation of a greenish layer of copper carbonate ($CuCO_3$) on its surface. This is called patination of copper.
4. Various methods such as galvanizing, tinning, alloying, etc. are used to protect metals from corrosion.

Q7 **Explain concept with example/explain with the help of a balanced equation.**
Minerals

TEXTBOOK Exercise · Q.4 (c) · Page 109

ANSWER / SOLUTION

1. Most metals being reactive do not occur in nature in the free state but are found in a combined state as their salts such as oxides, carbonates, sulphides, and nitrates.
2. However, some unreactive metals like silver, gold, platinum, generally occur in a free state.
3. The compounds of metals that occur in nature along with the impurities are called minerals.
4. For example, bauxite is a mineral of aluminium.

13 Draw a neat labelled diagram

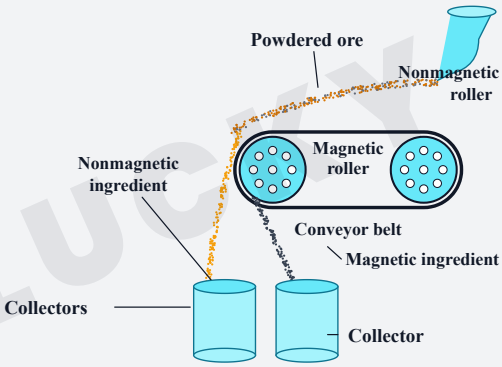
6 QUESTIONS · 4 ☐ Revised

Q1 Draw a neat labelled diagram.
Magnetic separation method.

TEXTBOOK Exercise · Q.8 (a) · Page 109

ANSWER / SOLUTION

Magnetic separation method-

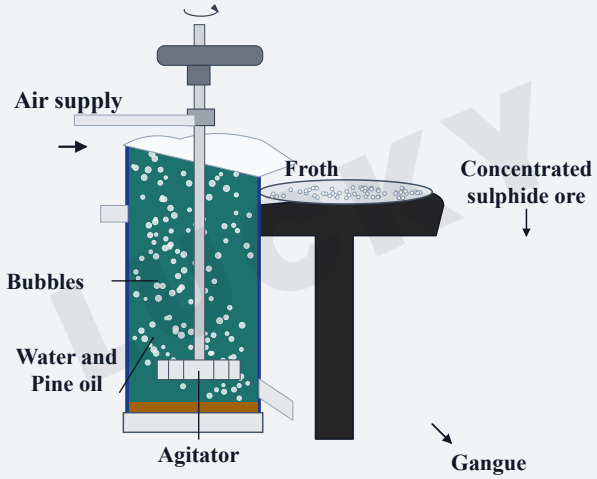


Q2 Draw a neat labelled diagram.
Froth floatation method.

TEXTBOOK Exercise · Q.8 (b) · Page 109

ANSWER / SOLUTION

Froth floatation method-

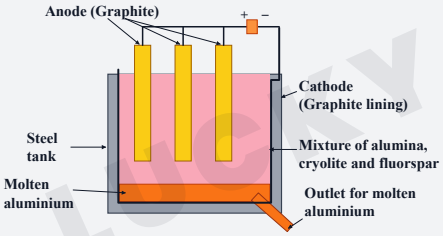


Q3 Draw a neat labelled diagram.
Electrolytic reduction of alumina.

TEXTBOOK Exercise · Q.8 (c) · Page 109

ANSWER / SOLUTION

Electrolytic reduction of alumina-

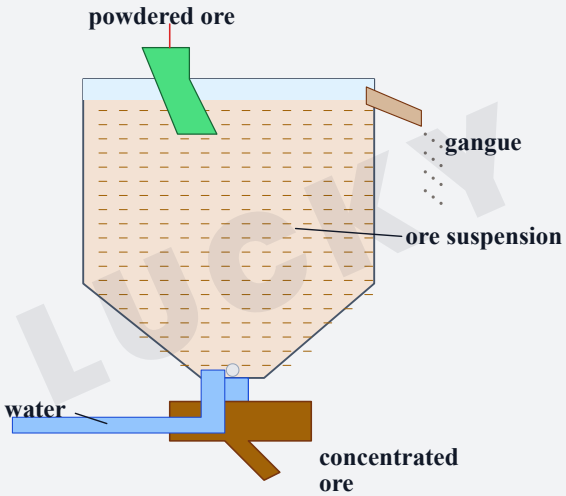


Q4 Draw a neat labelled diagram.
Hydraulic separation method.

TEXTBOOK Exercise · Q.8 (d) · Page 109

ANSWER / SOLUTION

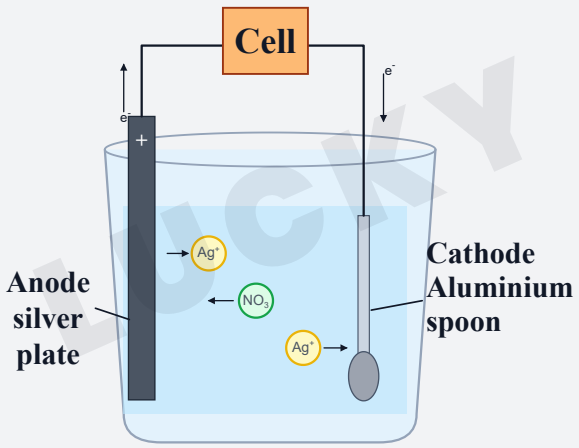
Hydraulic separation method-



Q5 Draw a neat labelled diagram.
Electroplating

ANSWER / SOLUTION

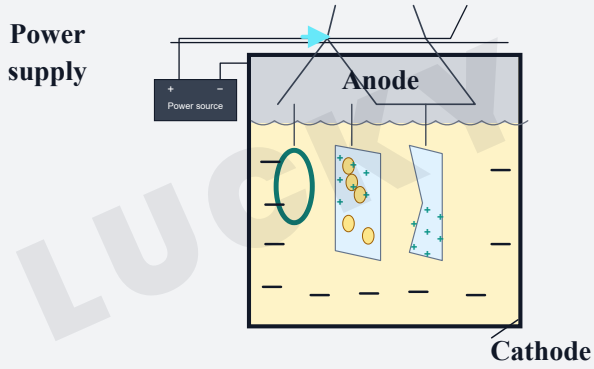
Electroplating-



Q6 Draw a neat labelled diagram.
Anodizing

ANSWER / SOLUTION

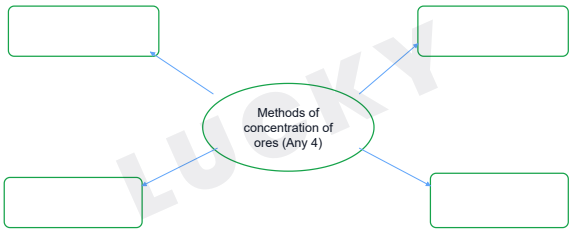
Anodizing-



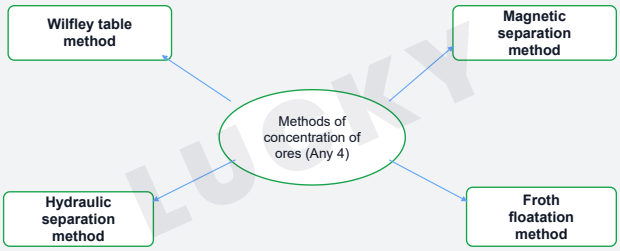
14 Complete flow chart given below

3 QUESTIONS ☐ Revised

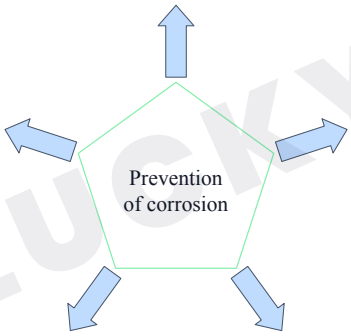
Q1 Complete flow chart given below.



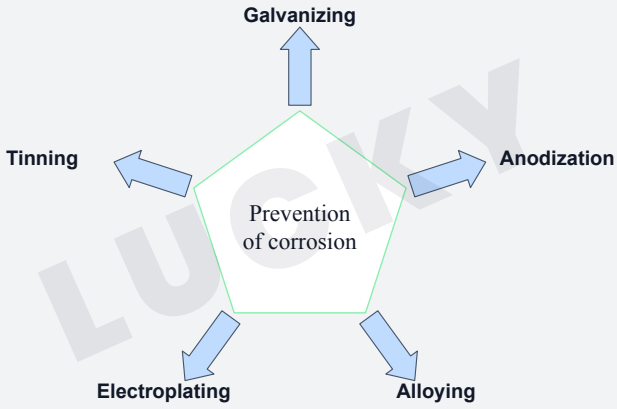
ANSWER / SOLUTION



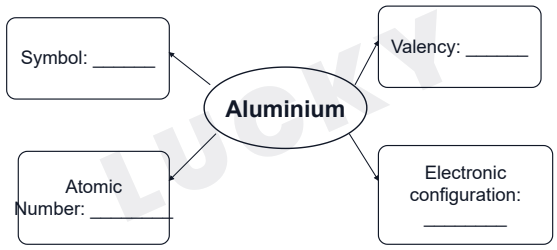
Q2 Complete flow chart given below.



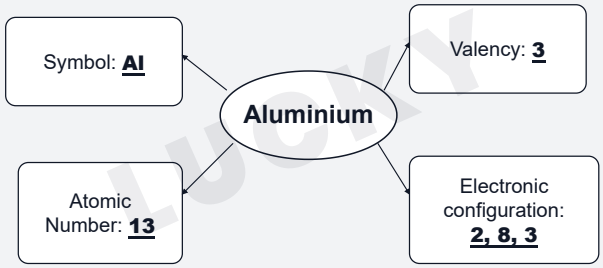
ANSWER / SOLUTION



Q3 **Complete flow chart given below.**



ANSWER / SOLUTION



15 Answer the Following

16 QUESTIONS ☐ Revised

Q1 **What is corrosion?**

ANSWER / SOLUTION

Corrosion is the damage caused to the metal by the chemical reaction of air, water, and acids with the surface of the metal. Generally, reactive metals corrode easily, and non-reactive metals have good resistance to corrosion.

Q2 **What is rust?**

ANSWER / SOLUTION

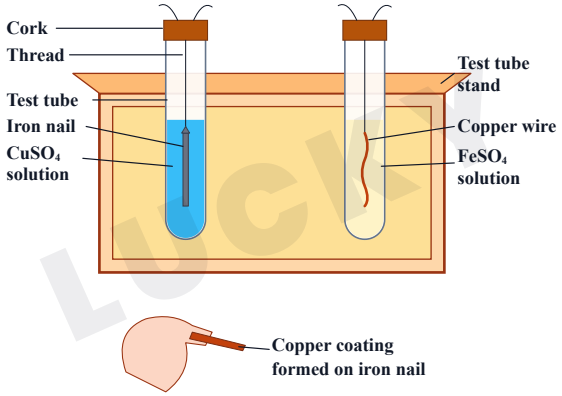
A certain type of reddish coloured solid layer collects on iron surface. This layer is called 'rust'.

Q3 **Write a molecular formula for rust.**

ANSWER / SOLUTION

The chemical formula of rust is $Fe_2O_3 \cdot xH_2O$.

Q4 **Observe the following diagram and identify the type of reaction and write observation.**



ANSWER / SOLUTION

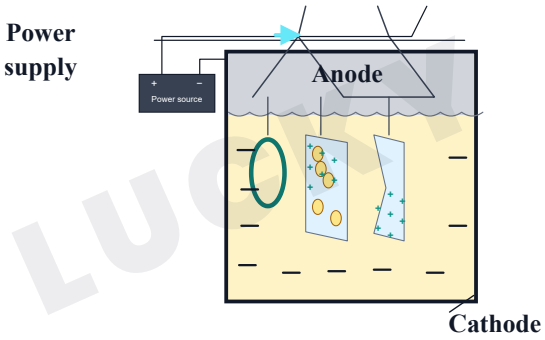
1. The type of reaction occurring in the given diagram is displacement reaction.

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

2. **Observations:**

- The blue colour of the solution will turn to green in test tube A and copper coating will form on iron nail because iron displaces copper from copper sulphate.
- The reaction will not occur in test tube B because iron is more reactive than copper.

Q5 **Observe the following diagram and give answers.**



1. **Name this method of prevention of corrosion.**
2. **For prevention of which metal this method is used?**
3. **What is used as anode in this method?**

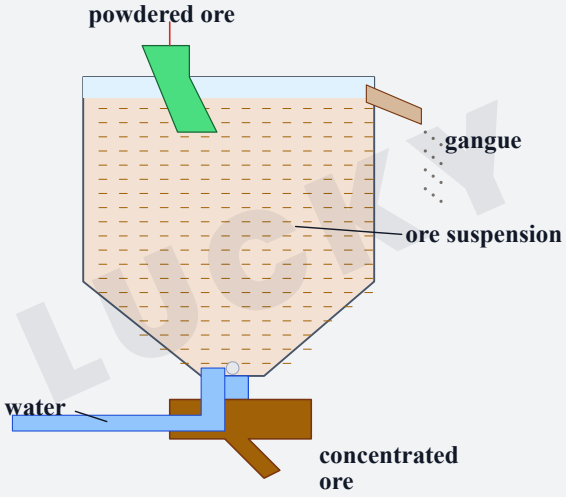
ANSWER / SOLUTION

1. The given diagram represents anodizing process used for the prevention of corrosion.
2. This method is used for the prevent corrosion of aluminium and copper.
3. In this method, copper or aluminium article is used as the anode.

Q6 **Explain the hydraulic separation method with a neat labelled diagram.**

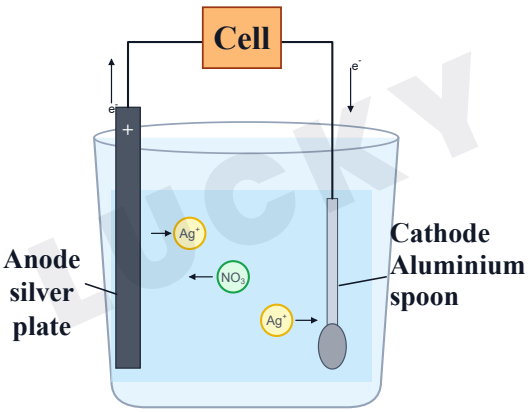
ANSWER / SOLUTION

1. This method is based on the working of a mill.
2. It has a tapering vessel similar to that used in grinding mill. The tapering vessel opens up in a tank-like container that is tapering on the lower side. The tank has a water inlet on the lower side and a water outlet on the upper side.
3. The powdered ore is released in the tank and forceful stream of water is introduced into the tank through the water inlet at the bottom.
4. The lighter particles of the gangue are carried up by the water jet and pass out along with water flowing out from the outlet near the top and get collected separately while the heavier ore particles are collected at the bottom from the lower side of the tank.
5. The reservoir is constructed in a conical shape so that the velocity of the water jet is reduced, thus preventing ore particles from being carried away along with the stream of water.
6. This method is based on the principle of gravitation law, wherein particles of identical size are separated by their weight with the help of water.



Hydraulic separation method

Q7 **Observe the following diagram and write answers.**

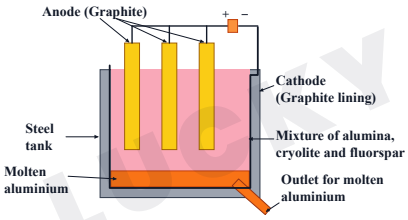


1. Name the method.
2. Explain the method.
3. Give two examples of this method.

ANSWER / SOLUTION

1. The method is called electroplating.
2. **Explanation:**
   a. Electroplating is the process in which a less reactive metal is coated on a more reactive metal by electrolysis.
   b. It protects the object made of reactive metal from corrosion.
   c. It also improves the appearance of the object.
3. Silver plated spoons, gold plated ornaments are examples of electroplating.

Q8 **Observe the following diagram and write answers.**



1. Name the method.
2. Write anode reaction and cathode reaction.
3. Why fluorspar and cryolite are added in the mixture?

ANSWER / SOLUTION

1. This method is electrolytic reduction of alumina.
2. **Anode reaction:**

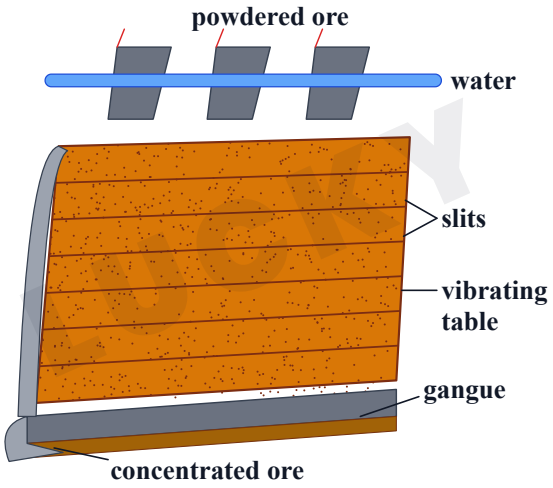
$$2O^{2-} \rightarrow O_2 + 4e^- \quad \text{(Oxidation)}$$

Cathode reaction:

$$Al^{3+} + 3e^- \rightarrow Al_{(l)} \quad \text{(Reduction)}$$

3. Alumina has a very high melting point ($> 2000\,^\circ C$). The purpose of mixing 'cryolite' and 'fluorspar' with 'alumina' in the electrolytic reduction of alumina is to carry out the process at a much lower temperature. Addition of cryolite and fluorspar reduces the melting point of alumina to about $1000\,^\circ C$.

Q9 **Identify the following method of concentration of ores and explain briefly.**

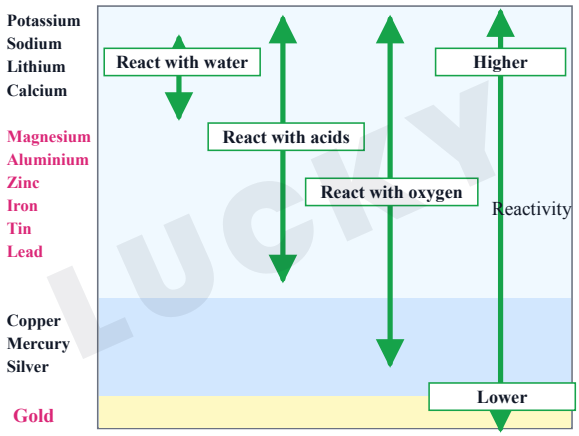


ANSWER / SOLUTION

1. This method represents Wilfley's table of concentration of ores.
2. **Explanation:**

- Wilfley's table is made by fixing narrow, thin and long wooden riffles on an inclined surface.
- The table is given a regular vibrating motion.
- The powdered ore obtained from lumps of the ore using ball mill is fed on the top of Wilfley's table and running stream of water is also released from the upper side.
- As a result, the lighter gangue particles are carried away by water and heavy ore particles in which proportion of minerals is more, settle between the wooden riffles fixed on the table and get collected on the slits between them.

Q10 **Observe the given figure of reactivity series of metals and answer the following questions:**



Reactivity series of metals

a. Name two metals which react with water.
b. Name two moderately reactive metals.
c. Name the most highly reactive metal and the most less reactive metal.

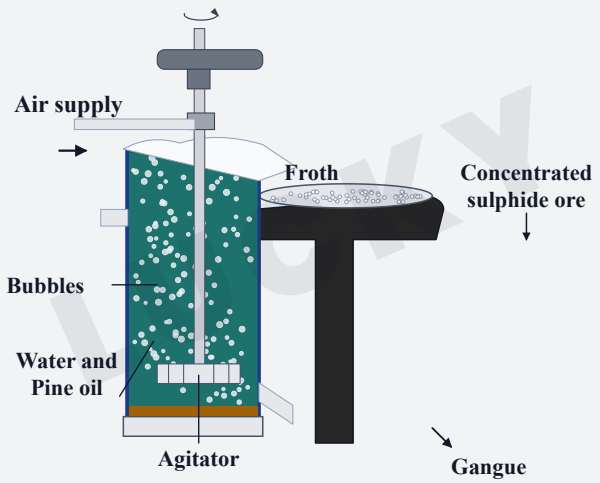
ANSWER / SOLUTION

a. Two metals that react with water: Potassium, sodium
b. Two moderately reactive metals: Zinc, iron
c. 1. Most highly reactive metal: Potassium
   2. Most less reactive metal: Gold

Q11 **Explain the froth floatation method with a neat labelled diagram.**

ANSWER / SOLUTION

1. The froth floatation process is based on two opposite properties of the particles, hydrophilic and hydrophobic.
2. The metal sulphide particles are hydrophobic. Due to this property, they get wetted mainly with oil. On the other hand, gangue particles are hydrophilic and get wetted by water.
3. In this method, the finely divided metal ore is added in a big tank containing large amount of water. To this, certain vegetable oil (pine oil, eucalyptus oil, etc.) is added for the formation of froth.
4. In the tank, compressed air is circulated through the water. Bubbles are formed due to this blown air. A rotating agitator at the centre of the floatation tank agitates the mixture and draws air into it to form bubbles. As a result, foam is formed which carries metal sulphide particles. The foam rises to the surface of water and floats. Therefore, this method is called froth floatation process.
5. This method is used to separate metal sulphide ores such as zinc blende (ZnS) and copper pyrite ($CuFeS_2$).



Froth floatation method

Q12 **Read the following passage and answer the questions.**

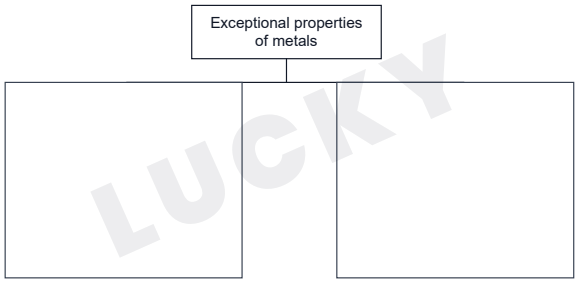
According to the reactivity series, zinc is more reactive than iron, iron is more reactive than silver. During study of this, a student dipped the iron nails in silver nitrate solution.

1. **What is reactivity series?**
2. **What will happen when iron nails are dipped in silver nitrate solution?**
3. **Which type of reaction happens when iron metal reacts with silver nitrate solution?**
4. **What will happen if a zinc rod is used instead of iron nail?**

ANSWER / SOLUTION

1. The arrangement of metals in the increasing or decreasing order of reactivity is called the reactivity series of metals.
2. Iron nails react with silver nitrate solution to form ferrous nitrate and silver. Silver nitrate solution is colourless while the colour of ferrous nitrate solution is brown. Hence, the solution will turn brown.
3. When an iron metal reacts with silver nitrate solution, a displacement reaction occurs.
4. Zinc rod reacts with silver nitrate solution to form zinc nitrate and silver.

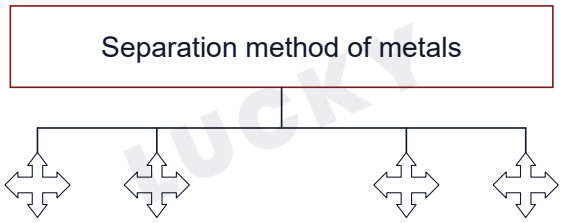
Q13 **Complete the following flowchart.**



ANSWER / SOLUTION

Exceptional properties of metals	
Properties	**Exceptions**
1. Physical state: Metals exist mainly in a solid state at room temperature. 2. Hardness: Metals are generally hard. 3. Melting and boiling point: Metals have high melting and boiling points.	1. Mercury and gallium exist in a liquid state at room temperature. 2. Lithium, sodium and potassium are metals which can be cut with a knife as they are very soft. 3. Sodium, potassium, mercury, and gallium have very low melting and boiling points.

Q14 **Complete the following flowchart.**



ANSWER / SOLUTION

Separation method of metals			
Wilfley table method	Hydraulic separation method	Magnetic separation method	Froth floatation method

Q15 **Complete the following flow chart.**



ANSWER / SOLUTION

Physical properties of metals				
Ductile and malleable	Good conductors of heat and electricity	Lustrous	Sonorous	Hard (except lithium, sodium, potassium)

Q16 **Write the properties of ionic compounds.**

ANSWER / SOLUTION

1. They are crystalline in nature.
2. They have strong forces of attraction between the ions. Therefore, the ionic compounds exist in solid state and are hard.
3. They are brittle and can be broken into pieces by applying pressure.
4. They have high melting and boiling points.
5. They are good conductors of electricity in the molten state and in aqueous solutions.
6. They are readily soluble in water and insoluble in solvents like kerosene, petrol, etc.

16 Answer the following questions

12 QUESTIONS ☐ Revised

Q1 **Explain in brief types of extraction of highly reactive metals according to their reactivity.**

ANSWER / SOLUTION

Extraction of highly reactive metals:

1. Highly reactive metals such as sodium, magnesium and calcium are obtained by electrolytic reduction.
2. The metals are deposited at the cathode (negatively charged electrode), whereas chlorine gas is liberated at the anode (positively charged electrode).
   E.g. For extraction of sodium metal, cathode and anode reactions are as follows:
   At cathode: $Na^+ + e^- \rightarrow Na$ (Reduction)
   At anode: $2Cl^- \rightarrow Cl_2 + 2e^-$ (Oxidation)

Q2 **Explain in brief types of extraction of moderately reactive metals according to their reactivity.**

ANSWER / SOLUTION

Extraction of moderately reactive metals:

1. Moderately reactive metals such as iron, zinc, lead, copper usually occur in the form of their sulphide salts or carbonate.
2. It is easier to obtain metals from their oxides rather than sulphides or carbonates.
3. Therefore, the sulphide ores are strongly heated in excess of air to transform them into oxides. This process is called roasting.
   E.g.
   $$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_2 \uparrow$$
4. Carbonate ores are strongly heated in a limited supply of air to transform them into oxides. This process is called calcination.
   E.g.
   $$ZnCO_{3(s)} \xrightarrow{\text{Heat}} ZnO_{(s)} + CO_2 \uparrow$$
5. The oxide so obtained is reduced to metal by using suitable reductant such as carbon.
   $$ZnO_{(s)} + C_{(s)} \xrightarrow{\text{Heat}} Zn_{(s)} + CO \uparrow$$
6. The metal thus obtained can be further refined by electrolysis.

Q3 **Explain in brief types of extraction of less reactive metals according to their reactivity.**

ANSWER / SOLUTION

Extraction of less reactive metals:

1. Less reactive metals such as gold, silver, platinum are found in a free state in nature.
2. However, the reserves of copper in the free state are very few. It is found mainly in the form of Cu_2S. Copper is obtained from Cu_2S ore just by heating in the air.
   $$2Cu_2S_{(s)} + 3O_{2(g)} \xrightarrow{\text{Heat}} 2Cu_2O_{(s)} + 2SO_2 \uparrow$$
   $$Cu_2S_{(s)} + 2Cu_2O_{(s)} \xrightarrow{\text{Heat}} 6Cu_{(s)} + SO_2 \uparrow$$
3. The copper metal thus obtained can be further refined electrolytically.

Q4 **Explain Bayer's process of concentration of bauxite with a chemical equation.**

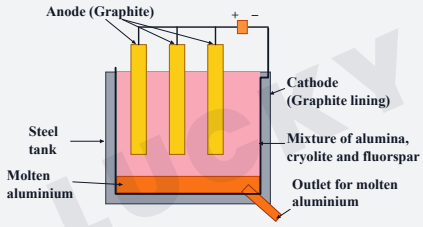
ANSWER / SOLUTION

1. Bauxite is the main ore of aluminium with silica (SiO_2), ferric oxide (Fe_2O_3) and titanium oxide (TiO_2) as impurities in it.
2. In this process, the bauxite ore is first crushed and then it is leached by heating with hot concentrated caustic soda ($NaOH$) solution under high pressure for 2 to 8 hrs at 140 to $150\,^\circ C$ in a tank called a digester.
3. Aluminium oxide, being amphoteric in nature, dissolves in an aqueous $NaOH$ solution, forming water-soluble sodium aluminate.
   $$Al_2O_3 \cdot 2H_2O_{(s)} + 2NaOH_{(aq)} \rightarrow 2NaAlO_{2(aq)} + 3H_2O_{(l)}$$
4. The iron oxide present in the gangue does not dissolve in an aqueous $NaOH$ solution and is separated by filtration.
5. However, silica from the gangue dissolves in an aqueous $NaOH$ solution forming water-soluble sodium silicate.
6. The solution of sodium aluminate is diluted with water and then cooled to $50\,^\circ C$. This gives aluminium hydroxide as a precipitate.
   $$NaAlO_{2(aq)} + 2H_2O_{(l)} \rightarrow NaOH_{(aq)} + Al(OH)_3 \downarrow$$
7. The aluminium hydroxide is then filtered, washed, dried and calcined by heating at $1000\,^\circ C$ to get pure aluminium oxide, called alumina.
   $$2Al(OH)_{3(s)} \xrightarrow[1000\,^\circ C]{\text{heat}} Al_2O_{3(s)} + 3H_2O_{(g)}$$

Q5 **Explain in brief electrolytic reduction of alumina with a neat labelled diagram.**

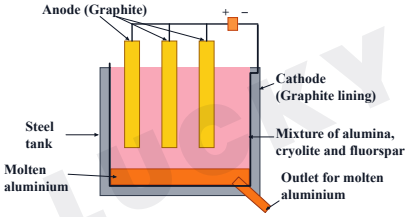
ANSWER / SOLUTION

1. The pure alumina obtained from bauxite ore by Bayer's process or Hall's process is electrolysed to obtain metallic aluminium.
2. Alumina has a very high melting point ($> 2000\,^{\circ}\mathrm{C}$).
3. Electrolysis of alumina is carried out in an electrolytic cell at a much lower temperature by dissolving it in molten cryolite ($\mathrm{Na_3AlF_6}$) and fluorspar ($\mathrm{CaF_2}$). Addition of cryolite and fluorspar reduces the melting point to about $1000\,^{\circ}\mathrm{C}$.



4. The electrolytic cell is a large steel tank having a lining of carbon (graphite) on the inner side. The carbon lining acts as a cathode. A set of carbon (graphite) rods dipped in the molten electrolyte acts as an anode.
5. On passing electric current, electrolysis of alumina takes place.
6. Aluminium is formed at the cathode and oxygen gas is liberated at the anode.
7. The electrode reactions are as follows:
   Cathode: $\mathrm{Al^{3+} + 3e^- \rightarrow Al_{(l)}}$ (Reduction)
   Anode: $\mathrm{2O^{2-} \rightarrow O_2 + 4e^-}$ (Oxidation)
8. The molten aluminium is heavier than the electrolyte used and hence, it sinks to the bottom of the tank from where it is removed periodically.
9. The oxygen gas liberated at the anode reacts with the carbon anode and forms carbon dioxide gas.
   $$\mathrm{C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}}$$
10. As the anode gets oxidized during the electrolysis of alumina, it has to be replaced from time to time.

Q6 **Observe the figure and answer the following.**

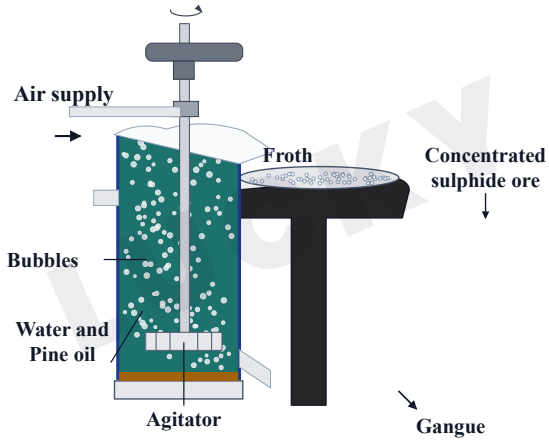


1. **Write the name of the method.**
2. **What is used as anode and cathode in this method?**
3. **Write the molecular formula and use of cryolite.**
4. **Write anode reaction.**
5. **Write cathode reaction.**

ANSWER / SOLUTION

1. The method shown is an electrolytic reduction of alumina.
2. A set of graphite rods dipped in the molten electrolyte acts as the anode, and the graphite lining acts as a cathode.
3. Molecular formula of cryolite: $\mathrm{Na_3AlF_6}$
   Use of cryolite: Addition of cryolite reduces the melting point of alumina to about $1000\,^{\circ}\mathrm{C}$.
4. Anode reaction: $\mathrm{2O^{2-} \rightarrow O_2 + 4e^-}$
5. Cathode reaction: $\mathrm{Al^{3+} + 3e^- \rightarrow Al_{(l)}}$

Q7 **Observe the figure and name and explain in brief the following method:**

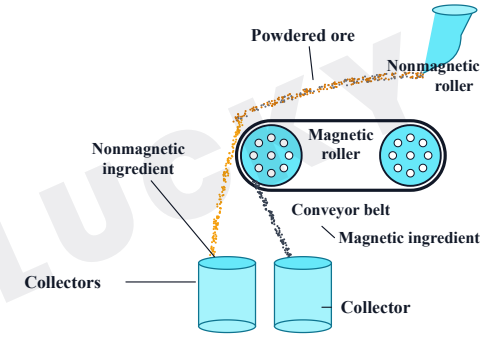


ANSWER / SOLUTION

The method shown in the figure is froth floatation method.

1. The froth floatation process is based on two opposite properties of the particles, hydrophilic and hydrophobic.
2. The metal sulphide particles are hydrophobic. Due to this property, they get wetted mainly with oil. On the other hand, gangue particles are hydrophilic and get wetted by water.
3. In this method, the finely divided metal ore is added in a big tank containing a large amount of water. To this, certain vegetable oil (pine oil, eucalyptus oil, etc.) is added for the formation of froth. A rotating agitator at the centre of the floatation tank agitates the mixture and draws air into it to form bubbles. As a result, foam is formed which carries metal sulphide particles. The foam rises to the surface of water and floats. Therefore, this method is called the froth floatation process.
4. This method is used to separate metal sulphide ores such as zinc blende (ZnS) and copper pyrite ($CuFeS_2$).

Q8 **Observe the figure and name and explain in brief the following method:**



ANSWER / SOLUTION

The method shown in the figure is the magnetic separation method.

1. The magnetic separation process is based on the differences in magnetic properties of the ore components.
2. This method requires an electromagnetic machine which consists of a nonmagnetic conveyor belt moving over two rollers, one of which is magnetic in nature while the other is nonmagnetic.
3. The powdered ore is dropped over the moving belt near the nonmagnetic roller. Two collector vessels are placed below the magnetic roller.
4. The nonmagnetic particles are carried further along with the belt and fall in the collector's vessel placed away from the magnetic roller. At the same time, the magnetic portion of the ore is attracted by the magnetic roller and falls in the collector vessel near the magnetic roller.

Q9 **Answer the questions on the following passage.**

The minerals from which the metal can be separated economically are called ores. Ores contain many types of impurities such as soil, sand and rocky substances along with metal compounds. These impurities are called gangue.

Metals can be extracted from their ores by means of various methods of separation. The process of extraction of metal in a pure state from the ores is also a part of metallurgy.

Ores are taken out from the mines and the gangue is usually separated from the ore at the site itself by various methods. Then the ores are carried out to the place where metals are produced. Here metals are extracted in pure form. Then metals are further purified by different methods of purification. This entire process is called metallurgy. Most metals being reactive do not occur in nature in the free state but are found in combined state as their salts such as oxides, carbonates, sulphides, and nitrates. however, the most unreactive metals that are not affected by air, water and other natural factors like silver, gold, platinum, generally occur in a free state. The compounds of metals that occur in nature along with the impurities are called minerals.

1. **What are ores?**
2. **Which processes are involved in the branch of metallurgy? What is metallurgy?**
3. **Which metals are found in a free state?**
4. **In what forms are metals found in combined state?**
5. **What is gangue?**

ANSWER / SOLUTION

1. The minerals from which the metal can be separated economically are called ores.
2. The extraction of metals from their ores by various methods of separation and further purification to obtain pure metals are involved in the branch of metallurgy.
   The extraction of metals from their ores and then refining them for use is known as metallurgy.
3. Metals such as silver, gold and platinum are found in free state.
4. Metals are found in combined states as their salts such as oxides, carbonates, sulphides, and nitrates.
5. The unwanted impurities of soil, sand, rocky substances, etc., present in the ore are called gangue.

Q10 **Explain the difference between Bayer's process and Hall's process by explaining the Bayer's process.**

ANSWER / SOLUTION

- **Difference between Bayer's process and Hall's process:**
  In Bayer's process, aluminium ore is heated with concentrated caustic soda while in Hall's process, it is heated with aqueous sodium carbonate to obtain water-soluble sodium aluminate. In Bayer's process, the solution of sodium aluminate is diluted with water to obtain aluminium hydroxide while in Hall's process, sodium aluminate is converted to aluminium hydroxide by reaction with CO_2.
- **Bayer's process:**

1. In this process, the bauxite ore is first crushed and then it is leached by heating with hot concentrated caustic soda ($NaOH$) solution under high pressure for 2 to 8 hrs at 140 to $150\,^\circ\text{C}$ in a tank called a digester.
2. Aluminium oxide, being amphoteric in nature, dissolves in an aqueous $NaOH$ solution, forming water-soluble sodium aluminate.
   $$Al_2O_3 \cdot 2H_2O_{(s)} + 2NaOH_{(aq)} \rightarrow 2NaAlO_{2(aq)} + 3H_2O_{(l)}$$
3. The iron oxide present in the gangue does not dissolve in an aqueous $NaOH$ solution and is separated by filtration. However, silica from the gangue dissolves in an aqueous $NaOH$ solution forming water-soluble sodium silicate.
4. The solution of sodium aluminate is diluted with water and then cooled to $50\,^\circ\text{C}$. This gives aluminium hydroxide as a precipitate.
   $$NaAlO_{2(aq)} + 2H_2O_{(l)} \rightarrow NaOH_{(aq)} + Al(OH)_3 \downarrow$$
5. The aluminium hydroxide is then filtered, washed, dried and calcined by heating at $1000\,^\circ\text{C}$ to get pure aluminium oxide, called alumina.
   $$2Al(OH)_{3(s)} \xrightarrow[1000\,^\circ\text{C}]{\text{heat}} Al_2O_{3(s)} + 3H_2O_{(g)}$$

Q11 **What is corrosion?**

ANSWER / SOLUTION

Corrosion is the damage caused to the metal by the chemical reaction of air, water, and acids with the surface of the metal. Generally, reactive metals corrode easily, and non-reactive metals have good resistance to corrosion.

Q12 **Give preventive methods by giving examples of corrosion?**

ANSWER / SOLUTION

Corrosion can be prevented by the following methods:

1. **Galvanizing:** Galvanizing is a process of giving a thin coating of zinc on iron or steel materials to protect them from corrosion. Zinc, being more electropositive than iron, undergoes corrosion first. Thus, it prevents corrosion of iron. However, if zinc layer goes away and the inner iron gets exposed, then the inner iron layer will start to rust.
2. **Tinning:** Tinning is the process of giving a coating of tin over other metals. Cooking vessels, made of copper and brass get a greenish coating due to corrosion. This greenish coating is poisonous. If food is kept in such a vessel, it gets spoiled. Therefore, copper and brass vessels are coated with tin (kalhai) to prevent corrosion.
3. **Anodizing:** The process of coating metals such as aluminium and copper with a thin and strong layer of their oxides by electrolysis is called anodizing process. For this, the copper or aluminium article is used as the anode. As this oxide layer is strong and uniform all over the surface, it is useful for the prevention of the corrosion of the metal.
4. **Electroplating:** Electroplating is the process in which a less reactive metal is coated on a more reactive metal by electrolysis. It protects the object made of reactive metal from corrosion. For example, silver-plated spoons, gold plated ornaments, etc.
5. **Alloying:** The homogenous mixture formed by mixing metal with other metals or nonmetals in a certain proportion is called an alloy. Alloying helps to decrease the intensity of corrosion of metals. For example, stainless steel does not get stains with air or water and also does not rust. It is an alloy made from 74% iron, 18% chromium and 8% carbon.

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